1. The `docs/bounds` frontier under procedure 2 (2026-10-23)

What this is. A scoreboard change, not a measurement: no session was run for it. The frontier page `docs/bounds/FRONTIER.md` moves from `ECFR1h8b45b672c283` (`ecbench.frontier/v1`, main `3a1ea541d`) to `ECFR1hbd02e945e867` (`ecbench.frontier/v2`) because the procedures that fit bounds and rank them changed (`docs/bounds/README.md` §3a, §5): the review in `DEEP_DIVE.md` §1.3 found that the v1 page published more than its data supported. All values come from committed sessions through `docs/bounds/refit.sh`; every published v1 record is unchanged and still re-derives (`ecbench bound check`), and each seed fit has a fit-v2 twin `<name>-v2.json`, which is what the v2 frontier ranks.

What changed in the procedure.

- Two bounds are compared only on the curves they share, per curve with that curve's interval; bounds sharing none are not comparable.
- Only fit-v2 bounds on four or more sizes of a whole session (no `--curve`, no `--tier`) are ranked; the rest, and every v1 record, are listed, not ranked.
- A method dominates only if every comparable bound of it does (ranked on its worst bound).
- Where either side has unpriced work, a dominance must hold at 0, 0.1 and 1 GAE per unpriced operation (AGENTS.md §5a); the break-even price is shown.
- A leader is named only when its intervals separate from every other method's on shared sizes; otherwise a tie set is listed.
- Fit v2: the exponent interval comes from resamples with at least three distinct sizes and exists only from four sizes; intervals that exclude their estimate, duplicate draws, unaudited or incomplete sessions and a non-default resample count are refused; tail diagnostics are recorded.

2. Per domain: old and new ranking

F frontier, D dominated, NR listed and not ranked (v1 records, fewer than four sizes, or a curve or tier subset).

domain	v1 F / D	v1 ops leader	v2 F / D / NR	v2 ops leader or tie set
koblitz toy	3 / 13	`calibration-koblitz-toy-rho-frob` (rho.signed_frobenius)	5 / 6 / 22	none; tie: `koblitz-bsgs-neg-v2` (bsgs.negation), `session-koblitz-toy-reference-e1-candidate-v2` (rho.signed_frobenius), `koblitz-rho-frob-v2` (rho.signed_frobenius)
koblitz medium	2 / 8	`pair-claw-koblitz-rho-strong` (rho.signed_frobenius_strong)	2 / 2 / 16	`pair-claw-koblitz-rho-strong-v2` (rho.signed_frobenius_strong)
prime toy	8 / 9	`calibration-prime-bsgs-neg` (bsgs.negation)	10 / 8 / 17	none; tie: `session-prime-toy-reference-e1-candidate-v2` (bsgs.negation), `calibration-prime-rho-neg-v2` (rho.negation), `session-prime-toy-reference-e1-incumbent-v2` (rho.negation), `prime-rho-neg-v2` (rho.negation), `calibration-prime-bsgs-neg-v2` (bsgs.negation), `prime-bsgs-neg-v2` (bsgs.negation)

The v1 leaders were named on point estimates. In prime toy, `bsgs.negation` (1.100 [0.978, 1.225]) and `rho.negation` (1.453 [1.184, 1.772]) overlap, so v2 names no leader. In Koblitz toy, the v1 leader was a three-size, tier-subset bound (`calibration-koblitz-toy-rho-frob`, degrees 17, 19, 23) compared with bounds fitted on other curves; it is no longer ranked, and on the four shared degrees the order between `rho.signed_frobenius` and `bsgs.negation` changes with the degree. In Koblitz medium the strong rho stays the ops leader, now because its intervals separate on all six shared degrees; the memory leader moves from a three-degree bound to the pair claw at the cryptanalysis#175 shape.



Figure 1: Ops ratio to the floor with 95 % two-stage bootstrap intervals for every fit-v2 bound, per domain; colour and marker give the v2 status, the label the status of the bound's published v1 record on the v1 page. Values from `records.json`; statuses from `frontier-v1.json` and `frontier-v2.json`.

3. Dominance claims that fail inside the pricing band

Each row is a pair the charged cost would rank but the band does not: the dominator wins at w = 0 and loses, or ties, by w = 1 GAE per unpriced operation. The break-even price is where the shared-size mean ops figures cross.

would be dominated	by	break-even (GAE / unpriced op)
koblitz-rho-plain-v2 (rho.plain)	koblitz-rho-frob-v2 (rho.signed_frobenius)	0.123
koblitz-kangaroo-v2 (kangaroo.vow)	koblitz-rho-frob-v2 (rho.signed_frobenius)	0.29
koblitz-kangaroo-v2 (kangaroo.vow)	koblitz-rho-neg-v2 (rho.negation)	0.417
calibration-prime-rho-plain-v2 (rho.plain)	calibration-prime-rho-neg-v2 (rho.negation)	0.397
calibration-prime-rho-plain-v2 (rho.plain)	prime-rho-neg-v2 (rho.negation)	0.337
calibration-prime-rho-plain-v2 (rho.plain)	session-prime-toy-reference-e1-incumbent-v2 (rho.negation)	0.404
prime-rho-plain-v2 (rho.plain)	calibration-prime-rho-neg-v2 (rho.negation)	0.55
prime-rho-plain-v2 (rho.plain)	prime-rho-neg-v2 (rho.negation)	0.419
prime-rho-plain-v2 (rho.plain)	session-prime-toy-reference-e1-incumbent-v2 (rho.negation)	0.491
calibration-prime-kangaroo-v2 (kangaroo.vow)	calibration-prime-rho-neg-v2 (rho.negation)	0.885
calibration-prime-kangaroo-v2 (kangaroo.vow)	prime-rho-neg-v2 (rho.negation)	0.975

4. Exponent intervals and tails

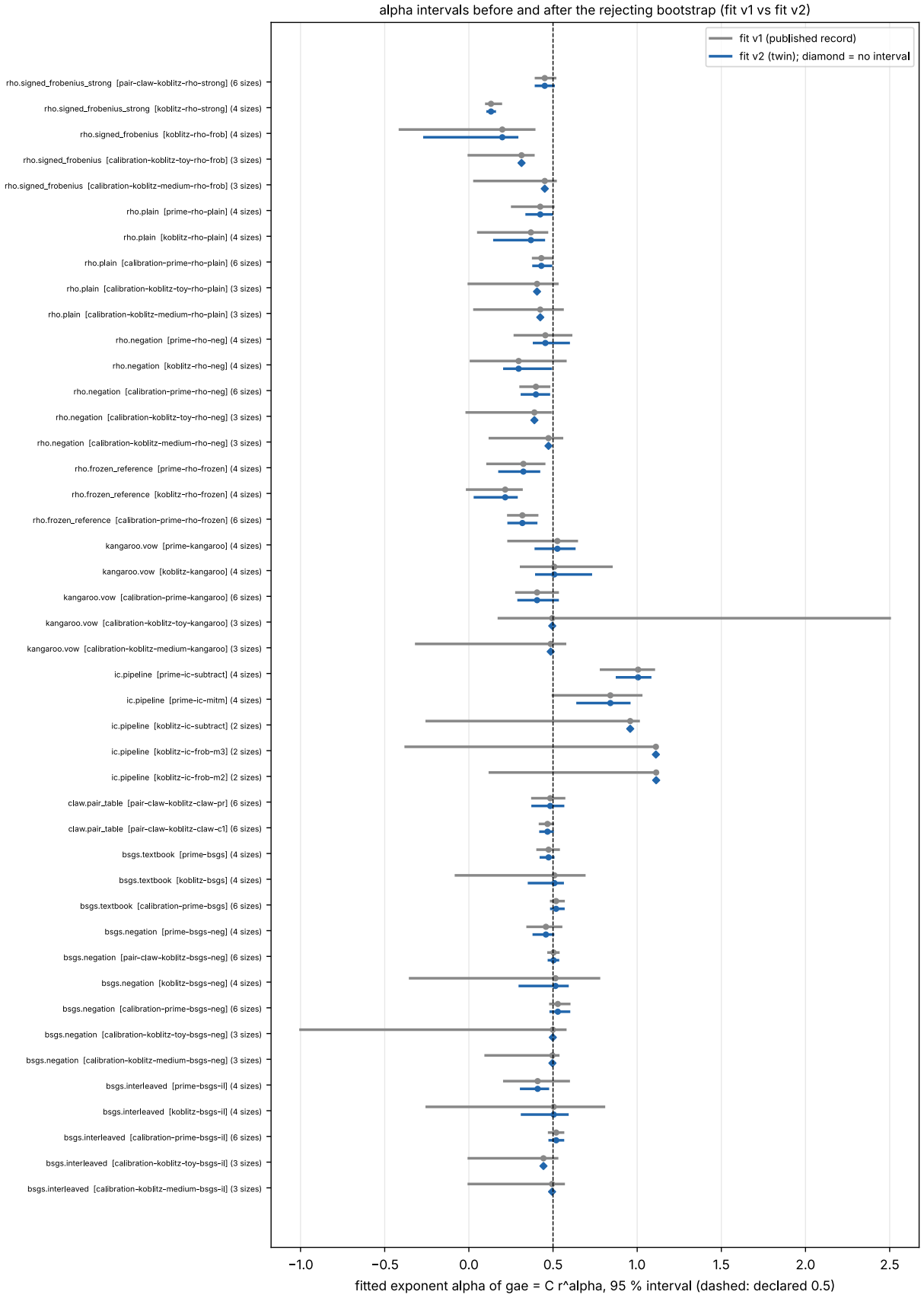


Figure 2: Fitted α with 95 % interval, published fit-v1 record (grey) and its fit-v2 twin (blue; a diamond means no interval: fewer than four sizes). The v1 ends at −1.0, 0.0, 2.5 and similar were degenerate one-size resamples.

bound (fit v2) with a large tail	runs	max / median	top-run share
`koblitz-rho-frob-v2`	96	54.24	0.245
`pair-claw-koblitz-claw-pr-v2`	144	10.286	0.043

prime-rho-frozen-v2	96	10.618	0.085
session-koblitz-toy-reference-e1-candidate-v2	96	11.672	0.094

5. Scope and limits

Toy and medium tiers only, planted targets, `ecbench.gae`. No bound here is a scaling result or a statement about cryptographic sizes. The v2 page is more conservative than v1: fewer bounds are ranked, ties stand, and an ordering that depends on how unpriced work is priced is left unresolved. The two epoch-1 challenge verdicts are procedure 1 and unchanged (`research/ecbench_bounds_challenges_20261006/`); their sessions' arms are ranked as ordinary fit-v2 bounds (`session-*-v2`). Canonical graphs checked: the frontier page and JSON (regenerated); `research/ecbench_frontier_20261013/` is a dated synthesis citing v1 ids that still exist, left unchanged; no other consumer of `docs/bounds/frontier.json` exists in `docs/`.